

Autonomous Navigation and Localization analysis in Industrial Environments for Quadruped Robots

Background

Quadruped robots are increasingly used for inspection tasks in industrial environments. Autonomous inspection requires the robot to respond to different environmental conditions, such as abnormal ambient data (e.g., noise, temperature), and adapt its behavior accordingly, including prolonged inspection at specific locations or repeated visits to critical areas. Such applications require flexible and robust navigation strategies that can operate safely in complex factory environments.

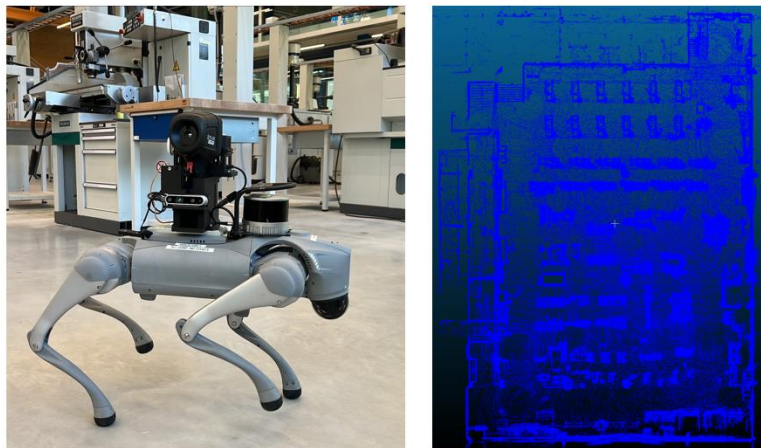


Fig. 1. a): robot dog is doing an inspection in the UT learning factory (Cube); b): SLAM

Objective

The objective of this master's assignment is to investigate autonomous navigation strategies for quadruped robots in industrial environments. The focus lies on developing flexible navigation methods ensuring safe operation in complex scenarios, with validation in a pilot factory.

Tasks

- Research on selecting suitable carrier / localization for autonomous factory inspection
- Investigate the localization accuracy from different SLAM algorithm
- Development of flexible independent navigation strategies for complex industrial scenarios.
- Integration with quadruped robot navigation user interfaces.
- Validation of navigation performance in industrial environments (e.g., Learning factory, Cube basement, and AMC Fraunhofer).

Contact

Dr. Poorya Ghafoorpoor Yazdi: p.ghafoorpooryazdi@utwente.nl

MSc. Leicai Xiao: l.xiao@utwente.nl